Ghulam Ishaq Khan Institute (GIKI)

Quiz # 1

Subject: Object Oriented Programming	**Course Code:** CS - 112 – Spring - 2025
Class: BS CY **Batch:** Fall – 2024	**Quiz Date:** 12 February
Course Instructor: Muhammad Ahmad Nawaz	**Total Marks:** 12
Course TA:	**Total Time:** 15 Minutes
Name:	Registration Number:

Note (Read notes & instructions first)

- First, read the instructions and statements of each question carefully.
  - Attempt all questions
  - Don't overwrite.
  - **CHEATING/COPY CASE will be graded as STRAIGHT ZERO MARKS.**

Sr. #	Answer	Statement

1.		Consider the following C++ code:
		`#include <iostream> #include <cstring>`
		`union Data { int i; float f; char str[20]; };`
		`int main() {`
		`Data data;`
		`data.i = 10;`
		`data.f = 220.5;`
		`strcpy(data.str, "C++ Programming");`
		`std::cout << "data.i : " << data.i << std::endl;`
		`std::cout << "data.f : " << data.f << std::endl;`
		`std::cout << "data.str : " << data.str << std::endl;`
		`return 0; }`
		
		What will be the output of this code?
		A. data.i : 10
		data.f : 220.500000
		data.str : C++ Programming
		B. data.i : 10
		data.f : 220.500000
		data.str : (garbage value or unpredictable)
		C. data.i : (garbage value or unpredictable)
		data.f : (garbage value or unpredictable)
		data.str : C++ Programming
		D. Compilation error due to incorrect union usage.

2.		Which of the following is a key characteristic of unions regarding memory usage? A. Unions allocate enough memory to store all their members simultaneously. B. Unions allocate memory only for the largest member. C. Unions dynamically allocate memory as needed for each member. D. The memory allocated to a union is fixed and cannot be changed during runtime.
3.		If an int occupies 4 bytes, what will be the output of the following C++ code?

```
#include <iostream>
union xyz {   char x, y, z, a, b, c, d, e;
  int i; };
int main() {   xyz my_union;
  std::cout << sizeof(my_union) << std::endl;
  return 0; }
```

A. 1
B. 4
C. 8
D. Undefined

4. Examine the following C++ code:

```
#include <iostream>
enum class Days {
   SUNDAY,
   MONDAY,
   TUESDAY,
   WEDNESDAY,
   THURSDAY,
   FRIDAY,
   SATURDAY };
int main() {
   Days today = Days::WEDNESDAY; // Scoped enum
   std::cout << "Day number: " << static_cast<int>(today) << std::endl;
   return 0; }
```

What will be the output of this program?

A. Day number: WEDNESDAY
B. Day number: 3
C. Day number: 4
D. Day number: 2

5 Which of the following is a benefit of using enums?

A. Enums allow for more efficient memory usage compared to integers.
B. Enums improve code readability and maintainability by using meaningful names instead of raw numbers.
C. Enums enable the creation of dynamic data structures.
D. Enums automatically handle type checking at runtime.

6 What will be the output of the following C code?

```
#include <iostream>
enum class MyEnum { a, b, c };
int main() {
  std::cout << sizeof(MyEnum) << std::endl;
  return 0; }
```

A. 3
B. Same as the size of an integer
C. 3 times the size of an integer
D. Error

7 What is a structure in C/C++?

A. A collection of variables of the same data type.
B. A user-defined data type that can group together variables of different data types under a single name.
C. A function that returns multiple values.

		D. A way to define constant values.

8		Consider the following C structure:
		`#include <iostream>`
		`struct Point {`
		`int x; int y; };`
		`int main() {`
		`Point p1 = {10, 20};`
		`Point *p2 = &p1;`
		`std::cout << "x = " << p2->x << ", y = " << p2->y << std::endl;`
		`return 0; }`
		What will the output of the code be?
		A. x = 0, y = 0
		B. x = 10, y = 20
		C. x = (garbage value), y = (garbage value)
		D. The code will not compile

9		Which operator is used to access members of a structure through a pointer? A. . (dot operator B. ->(arrow operator) C. :: (scope resolution operator) D. * (dereference operator)

10		Analyze the following C++ code:
		`#include <iostream>`
		`int factorial(int n) {`
		`if (n == 0) {`
		`return 1;`
		`} else {`
		`return n * factorial(n - 1);`
		`} }`
		`int main() {`
		`int num = 5;`
		`std::cout << "Factorial of " << num << " is " <<`
		`factorial(num) << std::endl;`
		`return 0;}`
		What will be the output of this program?
		A. Factorial of 5 is 5
		B.Factorial of 5 is 120
		C.Factorial of 5 is 0
		D. The program will result in a stack overflow error.

11		What is a crucial element to avoid infinite recursion? A. A return statement. B. A base case (a condition that stops the recursive calls). C. A loop within the recursive function. D. A global variable.

12		What will be the output of the following C code?
		`#include <iostream>`
		`int f1(int x) {`
		`int b;`
		`if (x == 1) return 1;`
		`else b = x * f1(x - 1);`
		`return b; }`
		`int main() {`
		`int n;`
		`n = f1(4);`
		`std::cout << n << std::endl;`
		`return 0; }`
		A. 24
		B. 4
		C. 12
		D. 10